ECET 250E-Linear Circuit 1
Problemset 2-Chapter 2- Basic Laws

Name: _____ Marks: _____

Chapter 2, Problem 2.

Find the hot resistance of a light bulb rated 60W, 120 V.

[Hint: $p = v^2/R \rightarrow \quad \mathbf{R} = v^2/p$]

Chapter 2, Problem 4.

(a) Calculate current *i* in Fig. 2.68 when the switch is in position 1.

(b) Find the current when the switch is in position 2.

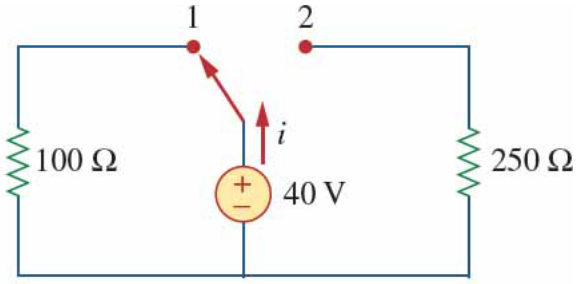


Figure 2.68
For Prob. 2.4.

Chapter 2, Problem 9.

Find i_1, i_2, and i_3 in Fig. 2.73.

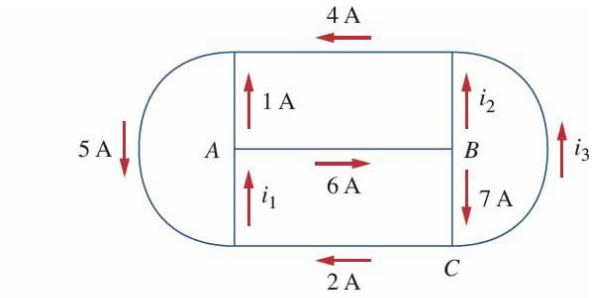


Figure 2.73
For Prob. 2.9.

Chapter 2, Problem 10.

Determine i_1 and i_2 in the circuit of Fig. 2.74.

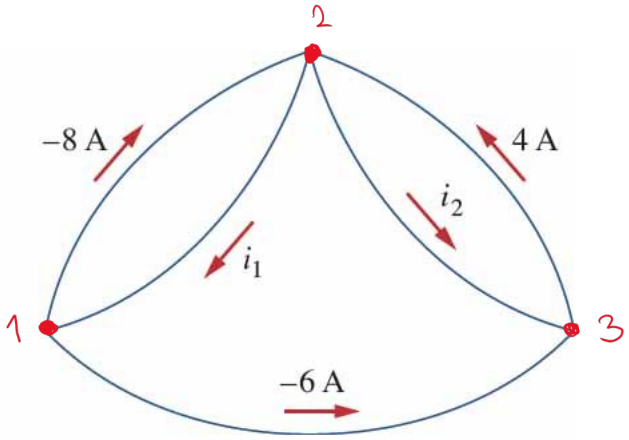


Figure 2.74
For Prob. 2.10.

Chapter 2, Problem 11.

In the circuit of Fig. 2.75, calculate V_1 and V_2.

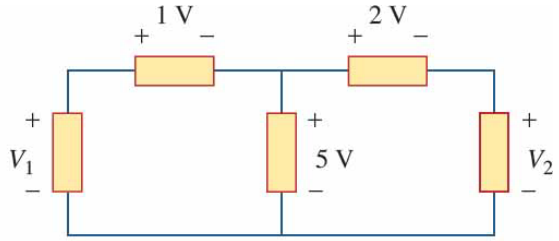


Figure 2.75
For Prob. 2.11.

Chapter 2, Problem 13.

For the circuit in Fig. 2.77, use KCL to find the branch currents I_1 to I_4.

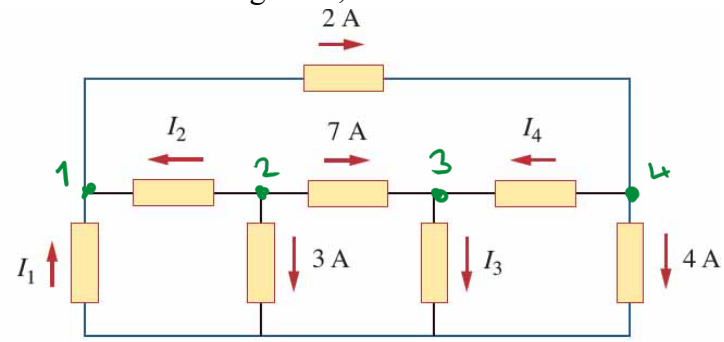


Figure 2.77
For Prob. 2.13.

Chapter 2, Problem 14.

Given the circuit in Fig. 2.78, use KVL to find the branch voltages V_1 to V_4.

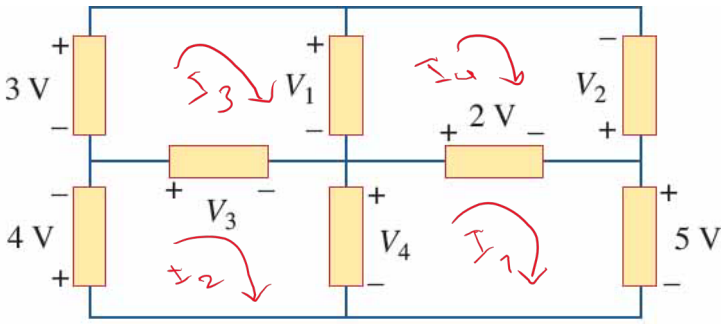


Figure 2.78
For Prob. 2.14.

Chapter 2, Problem 15.

Calculate v and i_x in the circuit of Fig. 2.79.

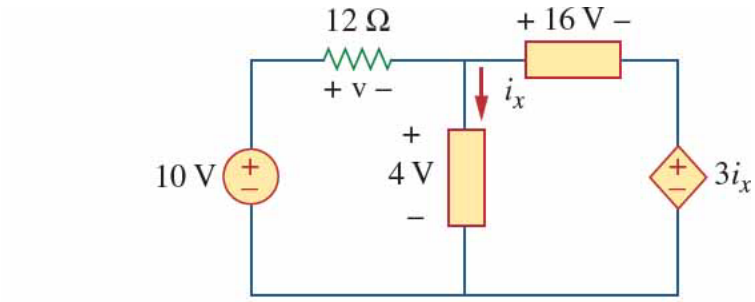


Figure 2.79
For Prob. 2.15.

Chapter 2, Problem 16.

Determine V_o in the circuit in Fig. 2.80.

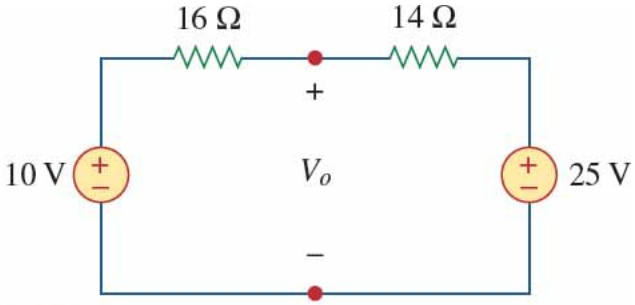


Figure 2.80
For Prob. 2.16.

Chapter 2, Problem 18.

Find I and V_{ab} in the circuit of Fig. 2.82.

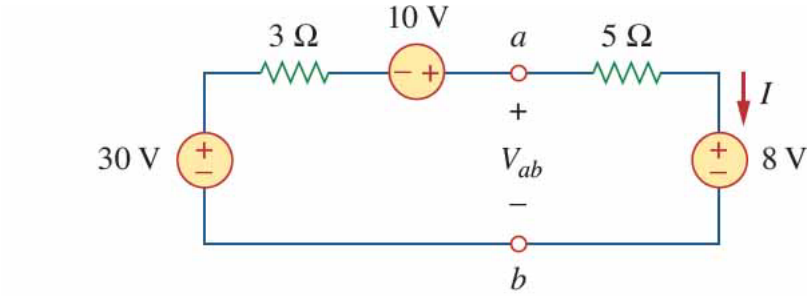


Figure 2.82
For Prob. 2.18.

Chapter 2, Problem 19.

From the circuit in Fig. 2.83, find *I*, the power dissipated by the resistor, and the power supplied by each source.

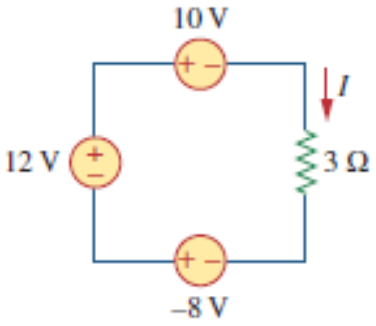


Figure 2.83
For Prob. 2.19.